

CUSTOMER CHURN PREDICTION FOR TELECOM RETENTION

**BUILDING A MACHINE LEARNING PIPELINE TO
REDUCE SUBSCRIBER LOSS**

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BUSINESS UNDERSTANDING

The Challenge	Telecom churn leads to significant revenue loss and high acquisition costs
Objective	Develop a classification model to predict which customers are likely to cancel their service.
Stakeholders	Customer Retention, Marketing, and Business Analysts.
Value	Enable proactive engagement (discounts, promotions) before the customer leaves.

SUCCESS METRICS (STRATEGY)

Primary Metric: Recall	Why? It is more expensive to miss a churner (lost revenue) than to send a discount to a loyal customer (small cost).
Precision & F1-Score:	To ensure the marketing budget isn't wasted on too many false alarms.
Accuracy	Used as a secondary baseline.

Understanding the Dataset

CONTENT:

- 3333 CUSTOMERS, 21 FEATURES
- INCLUDES USAGE, CHARGES, AND SERVICE CALLS
- TARGET VARIABLE: CHURN

The heatmap shows relationships between features, helping us understand which variables are related to churn behavior.

DATA PREPARATION WORKFLOW

Cleaning	Dropping unique identifiers (Phone Number).
Encoding	Converting "Yes/No" plans into binary (0 and 1).
Splitting	80/20 Train-Test split to ensure the model
Scaling	Standardizing numerical features so usage minutes and call counts are on the same scale.

MODELLING AND EVALUATION

Decision Tree (Better at capturing complex customer behaviors).

Logistic Regression (Simple and interpretable).

- **Interpretation:** Identifying which factors (like high customer service calls) most influence the decision to leave.
- **Visual Suggestion:** A placeholder for your Confusion Matrix (once you run the model).

Customer Churn Prediction in Telecom

CONTENT:

- TELECOM COMPANIES LOSE CUSTOMERS TO COMPETITORS (CHURN)
- RETAINING CUSTOMERS IS CHEAPER THAN ACQUIRING NEW ONES
- GOAL: PREDICT CUSTOMERS LIKELY TO CHURN

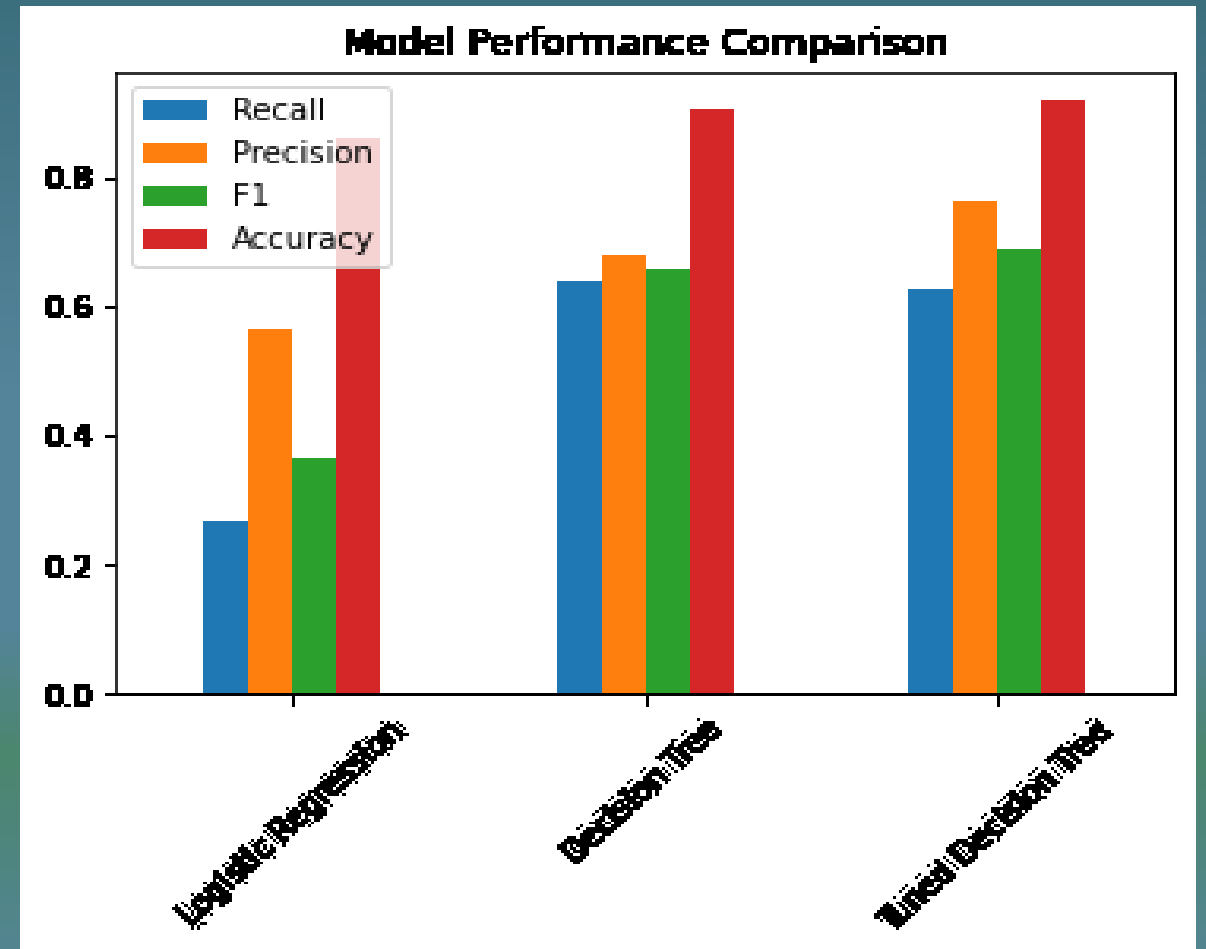
The dataset shows class imbalance, with far more non-churning customers than churners. This impacts how we evaluate models.

Machine Learning Models Built

CONTENT:

- LOGISTIC REGRESSION (BASELINE)
- DECISION TREE
- TUNED DECISION TREE (FINAL MODEL)

We compared multiple models and selected the tuned decision tree based on recall performance.

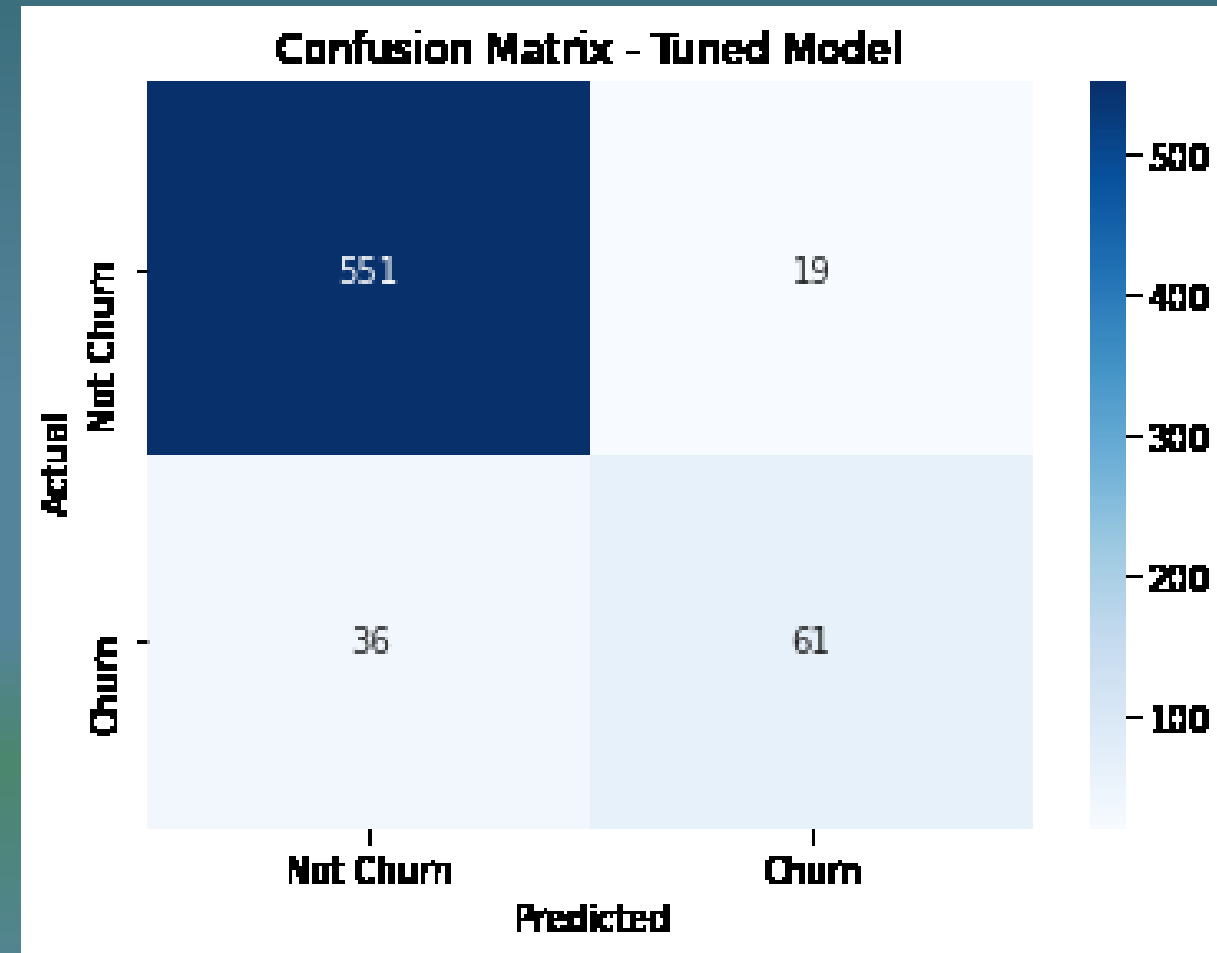


Model Evaluation (Confusion Matrix)

CONTENT:

- RECALL IS THE MOST IMPORTANT METRIC
- WE FOCUS ON DETECTING CHURNERS CORRECTLY

The confusion matrix shows that the tuned model correctly identifies most churners, reducing missed high-risk customers.



Key Drivers of Customer Churn

CONTENT:

- CUSTOMER SERVICE CALLS
- HIGH CHARGES
- USAGE PATTERNS

The most important factors influencing churn are customer service calls and high usage charges, which can guide business retention strategies.

BUSINESS CONSTRAINTS & RECOMMENDATIONS

- Dataset lacks external competitor pricing and real-time behavioral shifts.
- Actions:
 1. Proactively contact users with >3 service calls.
 2. Review the value proposition of the "International Plan."
 3. Use model probabilities to rank customers by "Risk Score."